FILED BY GUATEMALA'S 18.4 MILLION RESIDENTS IN 2024. NOT EIGHT THOUSAND.

Eight patents

What Mexico and Central America stopped buying, and what it cost them

Growth diagnostic · one subregion, different binding constraints

Group 1 · [Firm name] · IE University
[Presenter 1] · [Presenter 2] · [Presenter 3] · [Presenter 4] · [Presenter 5]

One subregion, different binding constraints.

GROUP	BINDING CONSTRAINT	WHAT FITS
Mexico	Missing supplier linkages and the informality wedge	**Linkages and formalisation**
Northern Triangle and Nicaragua	State capacity and violence; people are the main export	**Revenue, security and connectivity**
Costa Rica	Contrast: grew on FDI plus skills, without R&D or patents	**The model for capability-building**
Panama	Contrast: fast growth on services, low tax, high informality	**Growth that was not inclusive**

Evidence on slides 3 to 14 · programmes on slides 15 to 17

Seven economies, four different stories.

SEVEN ECONOMIES AGAINST THE OECD

	Informal jobs %	Tax % of GDP	Remittances % GDP	PISA maths
Mexico	55.5	17.7	3.7	395
Guatemala	83.2	14.0	19.1	344
Honduras	82.6	21.0	25.7	n.t.
El Salvador	66.5	22.8	24.3	343
Nicaragua	63.0	27.5	26.6	n.t.
Costa Rica	37.4	24.9	0.7	385
Panama	58.7	11.9	0.6	357
OECD		**34.0**		**472**

~25%

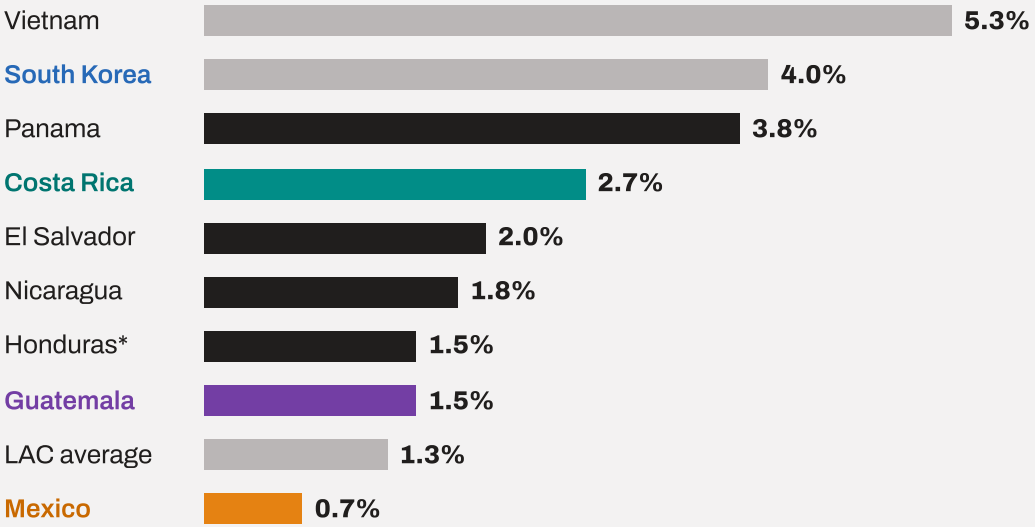
of GDP is money sent home, in Honduras, El Salvador and Nicaragua

Darker cells sit further behind the Latin American average. Costa Rica and Panama break the pattern in different ways.

ILOSTAT · Revenue Statistics in LAC 2025 (2023 data) · WDI remittances 2024 · OECD PISA 2022

Since 1990, Mexico has grown slowest of all.

REAL INCOME PER PERSON, AVERAGE ANNUAL GROWTH, 1990 TO 2024



0.7%

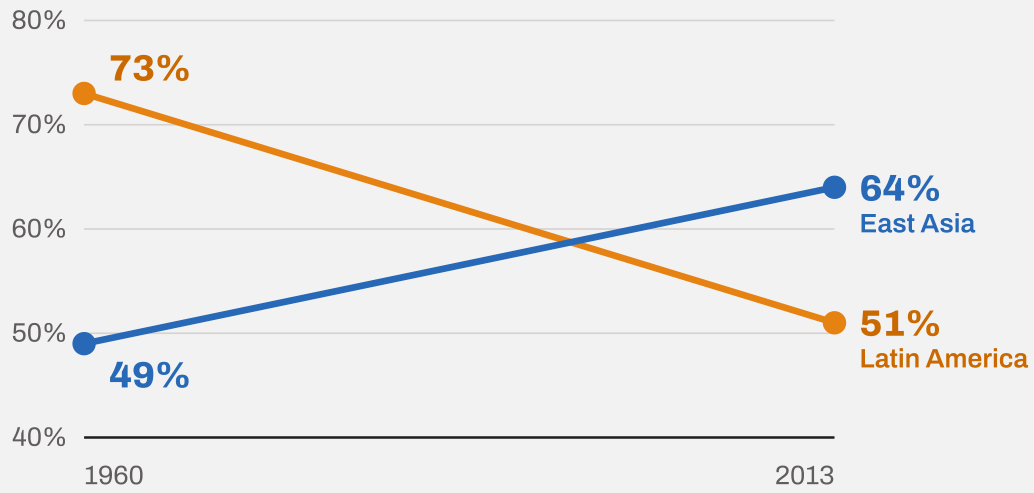
a year: Mexico's growth in income per person since 1990. Vietnam: 5.3%.

Panama grew fastest in the subregion, on services. Costa Rica came second.

World Bank WDI, NY.GDP.PCAP.KD, 1990 to 2024, compound annual rate · *Honduras from 2000 (series break) · grey: benchmarks

Latin America did not lack machines. It used them less well.

TOTAL FACTOR PRODUCTIVITY, % OF THE US LEVEL



54%

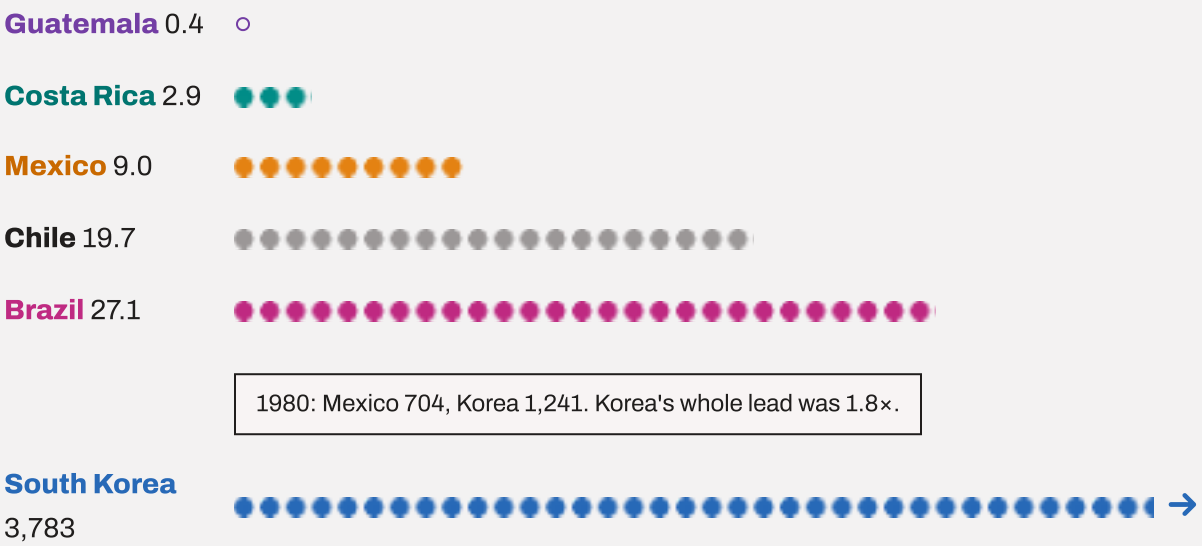
richer, had productivity simply kept pace with the US

Chile is the only country in the region that closed any of the gap.

Grazzi and Pietrobelli (IDB, 2016) · Pagés (IDB, 2010) · Penn World Table 10.01

Patents tell the same story, more starkly.

RESIDENT PATENTS PER MILLION PEOPLE, 2024



420×

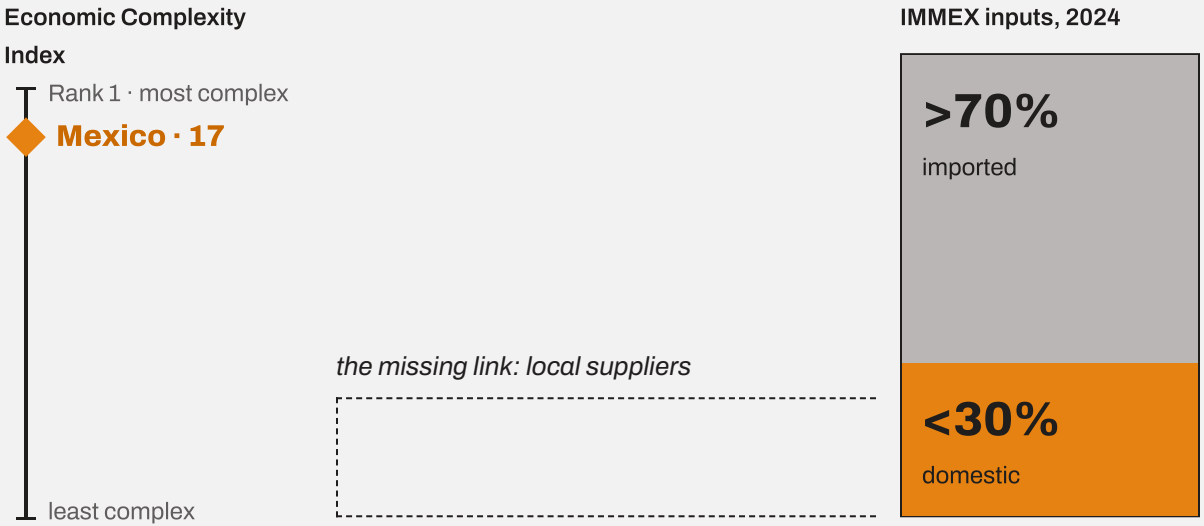
Korea's resident patents per person against Mexico's, 2024. Against Guatemala: 8,704×.

Patents undercount invention, since multinationals file abroad. We use them as one signal, not the measure.

World Bank WDI, IP.PAT.RESD and SP.POP.TOTL, 2024 · retrieved 17 Oct 2026

The complexity sits in the product. The capability sits abroad.

WHAT MEXICO EXPORTS, AND WHAT IT BUYS TO MAKE IT



>70%

of the inputs used by Mexico's export factories are imported, every year since 2013

a factory that imports its parts exports wages, not know-how

Harvard Growth Lab, Atlas of Economic Complexity · INEGI IMMEX, 2024

Same test, different verdicts for Mexico and Guatemala.

CONSTRAINT	INDICATOR	MEXICO	GUATEMALA	LAC
Finance	Private credit, % of GDP · lending spread, points	**35.5 · 6.2** Does not bind	**37.2 · 7.7** Does not bind	52.9 · 6.1
Human capital	PISA 2022 maths score (OECD 472)	**395** Partly binds	**344** Binds	n/a
Infrastructure	Logistics index, 1 to 5 · electricity access, %	**2.9 · 99.8** Does not bind	**2.6 · 90.9** Binds	2.7 · 98.0
Appropriability	Informal jobs % · homicides per 100k · tax % of GDP	**55.5 · 19.3 · 17.7** Binds	**83.2 · 16.1 · 14.0** Binds	n/a · 20.2 · 21.3

1 vs 3

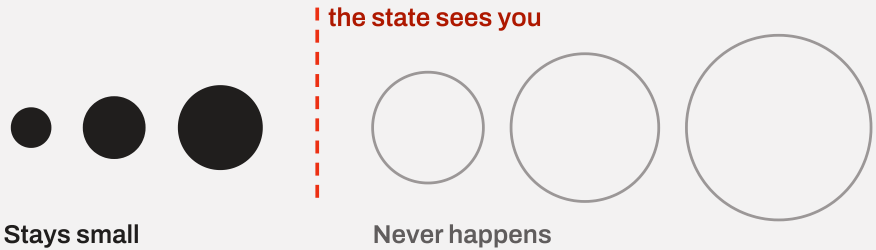
constraints that clearly bind, in Mexico against Guatemala. Finance binds in neither.

Mexico's binding constraint is appropriability: informality and missing supplier links. Guatemala's is the state itself.

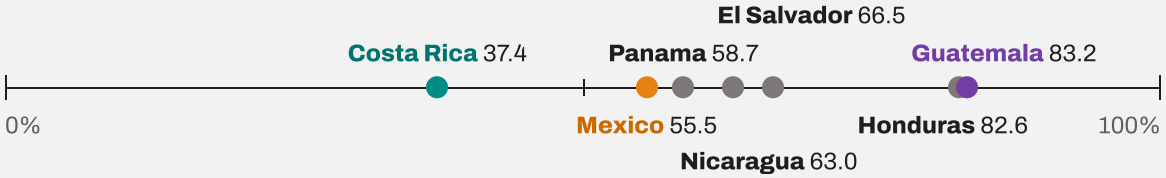
Framework after Hausmann, Rodrik and Velasco (2005) · WDI FS.AST.PRVT.GD.ZS, FR.INR.LNDP 2025; LP.LPI.OVRL.XQ 2022; EG.ELC.ACCS.ZS 2024 · OECD PISA 2022 · ILOSTAT · InSight Crime 2024 · Revenue Statistics in LAC 2025

In Mexico, every formal hire carries a tax, so firms stay small.

WHY FIRMS NEVER GET BIG



Informal share of workers, ILO harmonised



55%

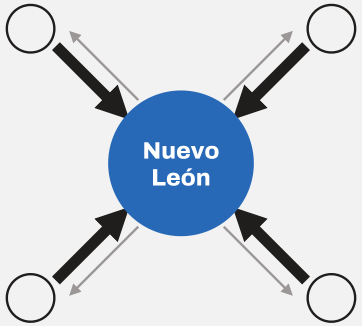
of Mexican workers are informal. Costa Rica: 37%. Guatemala: 83%.

Levy puts the efficiency cost at 0.9% to 1.44% of GDP every year.

ILOSTAT, latest year · mechanism after Levy (2008)

The export poles buy abroad, and the south is left out.

THE GROWTH POLE, IN PRACTICE



Nuevo León's export factories buy 39% of all domestic inputs purchased by Mexico's IMMEX firms.

Multidimensional poverty, % of people, 2024

Chiapas · southern border



Mexico · national



Nuevo León · industrial north



6×

the poverty rate in Chiapas (66%) against Nuevo León (11%), 2024

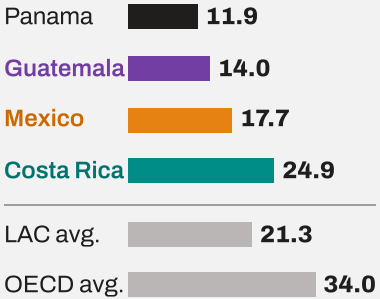
Even the local supplier links that exist cluster inside the pole: Nuevo León's factories make 39% of IMMEX domestic purchases.

INEGI poverty 2024 · El Financiero (2023) on IMMEX · Perroux, Myrdal, Hirschman

In the Northern Triangle, the state is too thin to provide the basics.

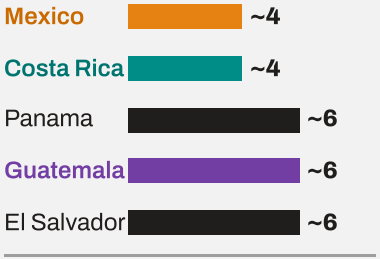
REVENUE

Tax revenue, % of GDP, 2023



SKILLS

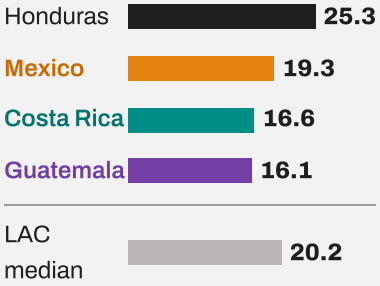
Years of schooling behind the OECD, PISA 2022 maths



20 PISA points equal about one school year. OECD mean: 472.

SAFETY

Homicides per 100,000 people, 2024



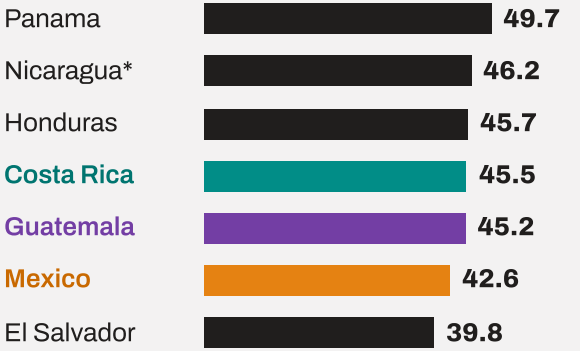
14%

Guatemala's tax take, % of GDP. Latin America: 21.3%. OECD: 34%.

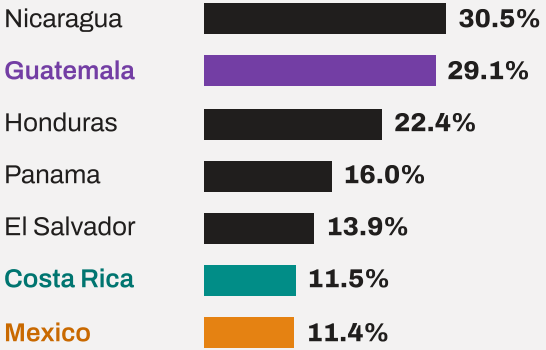
Violence works as a second, private tax on growth, and it falls hardest on the periphery.

Growth has not been shared, and the poorest farm on rain.

INCOME INEQUALITY, GINI INDEX



JOBS IN AGRICULTURE, % OF EMPLOYMENT



Guatemala, Honduras, El Salvador and Nicaragua share the Dry Corridor, where most farming depends on rain and droughts recur.

~30%

of jobs in Guatemala and Nicaragua are in farming, exposed to drought in the Dry Corridor

has anyone in or near the region broken these walls?

WDI SI.POV.GINI, latest year 2023 to 2025 (*Nicaragua 2014) · WDI SL.AGR.EMPL.ZS, 2025 (ILO modelled) · Dry Corridor: FAO and WFP regional assessments

Costa Rica grew on skills. Panama grew without inclusion.

	COSTA RICA	PANAMA
Growth a year since 1990	**2.7%**	**3.8%**
R&D, % of GDP	**0.34**	**0.20**
Tax, % of GDP	**24.9**	**11.9**
Informal jobs	**37%**	**59%**
Gini index	**45.5**	**49.7**

44%

of Costa Rica's goods exports in 2024 were medical devices, with almost no R&D or patents

Panama grew fastest, on canal and financial services, but collects the least tax and leaves most workers informal.

Each constraint has already been broken, close to home.

4 → 29

Vietnam · supplier development, 2014 to 2017

Samsung's local tier-1 suppliers rose sevenfold in three years.

Mexico

–90%

Medellín · social urbanism, 2004 onward

Transport and public buildings went to the poorest comunas first. Homicides fell about 90% from 1991.

Northern Triangle

44%

Costa Rica · CINDE, 1982

Targeted investment promotion plus skills: medical devices reached 44% of goods exports.

Capability

3

precedents, one for each programme, each a narrow mandate held for a long time

A caution: Costa Rica cut CINDE's public funding in 2023. These institutions need a protected mandate.

Mexico: link the export plants to local firms, and lift the payroll wedge.

LINKAGES: PAY FOR THE TWO STEPS MARKETS SKIP

01	02 · fund	03	04	05 · fund
A big factory arrives	**Local firms learn to supply it**	Skills spread	It runs on its own	**Roads reach the towns outside**

Formalisation: move health and pensions from wages to general revenue, so firms can grow into suppliers. Modelled net cost: 0.34% of GDP.

PILOT

Nuevo León pole to Chiapas

IDB INSTRUMENT

Investment loan, IDB Invest supply-chain finance, then a policy-based loan

MEASURED BY

Domestic share of export inputs, now under 30%

<30%

domestic share of inputs in Mexico's export factories: the number this programme moves

Perroux's growth pole · Vietnam supplier development (UNIDO 2019) · Antón, Hernández and Levy (2012), IDB · IDB border integration programme

Northern Triangle: build the state first, then connect the towns.

REVENUE

14.0%

tax take, Guatemala, 2023

Broader VAT base and e-invoicing

SECURITY

25.3

homicides per 100k, Honduras, 2024

A security component in every corridor

CONNECTIVITY

88.7%

electricity access, Nicaragua, 2024

Roads, power and broadband to lagging towns

PILOT

Guatemala first

IDB INSTRUMENT

Policy-based loan for revenue, investment loans for corridors

MEASURED BY

Tax take, homicides, travel time to the nearest town

14%

Guatemala's tax take. Every other fix is paid for from this base.

Revenue Statistics in LAC 2025 · InSight Crime 2024 · WDI EG.ELC.ACCS.ZS · precedent: Medellín social urbanism

Across the region: an agency that proves new industries, then leaves.

WHO CARRIES THE RISK, OVER TIME



Agency proves it → **Hands it to firms** → **Agency leaves**

MODEL

CINDE, Costa Rica, 1982

PILOT

One Mexican export state, then the isthmus

MEASURED BY

Two new export products by year five

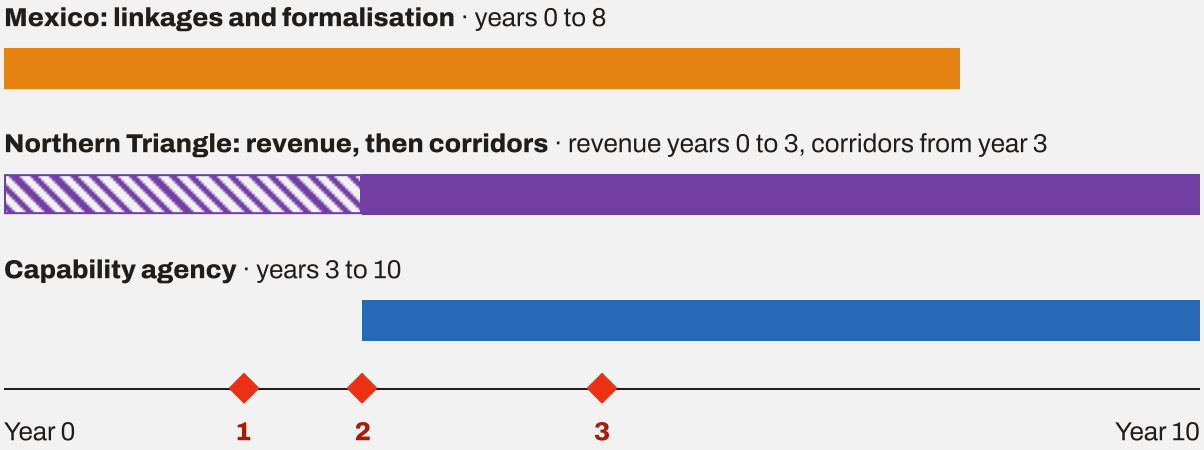
US$35m

the IDB loan that funded Costa Rica's innovation and skills programme (CR-L1043, 2012): our sizing benchmark

Vernon 1966 · Hausmann and Rodrik 2003 · precedents: CINDE Costa Rica, Fundación Chile · IDB project CR-L1043

The cheap, visible programme goes first.

TEN YEARS, THREE PROGRAMMES



1 · Year 2 first certified local suppliers

2 · Year 3 Guatemala's tax up one point

3 · Year 5 first new export product

2

years to the first visible result: local suppliers certified in Mexico

Each programme earns the credibility the next one spends.

Timeline illustrative · gates to be calibrated with the IDB country teams

We promise three counts, not a growth rate.

THE THREE COUNTS, WITH BASELINES

Linkages · Mexico

●○○○○

<30%

domestic share of export-factory inputs, 2024

Against Vietnam's 20% to 25% localisation

Formal jobs · Mexico

●●○○○

55.5%

of workers informal

Against Costa Rica, 37.4%

State capacity · Guatemala

●○○○○

14.0%

tax take, % of GDP, 2023

Against the LAC average, 21.3%

Top risk: a US trade or remittance shock. Programme three also targets non-US buyers.

3

counts, published every year, each with a baseline and a fair comparator

Baselines: INEGI IMMEX, ILOSTAT, Revenue Statistics in LAC 2025 · targets set with the IDB country teams

Fund linkages and formalisation. In Mexico. Starting in 2027.

Record investment is landing in Mexico now. Linkages decide whether it builds capability or only assembly lines.

Months 0 to 3
Supplier scan in two corridors

Months 3 to 9
Design the supplier programme, prepare the loan

Months 9 to 12
Loan approval; formalisation groundwork

sixty-four years is long enough to wait for the second attempt

APPENDIX

Backup

A1 to A4 · the evidence behind the story and answers for Q&A

Every indicator, every country

Indicator	Mexico	Guatemala	Honduras	El Salvador	Nicaragua	Costa Rica	Panama	Year · source
GDP per head, PPP US$	25,820	14,393	7,503	13,258	8,721	31,552	41,430	2024 · WDI
Resident patent filings	1,172	8	8 (2018)	6	1 (2014)	15	11	2024 · WDI
R&D spending, % of GDP	0.25	0.01 (2023)	0.06 (2019)	0.10 (2023)	0.11 (2015)	0.34 (2023)	0.20	2024 · WDI
Researchers per million	438	1.4 (2023)	188 (2019)	55 (2023)	71 (1997)	508 (2023)	250	2024 · WDI
Informal jobs, % of workers	55.5	83.2	82.6	66.5	63.0	37.4	58.7	Latest · ILOSTAT
Tax revenue, % of GDP	17.7	14.0	21.0	22.8	27.5	24.9	11.9	2023 · Rev. Stats LAC
Remittances, % of GDP	3.7	19.1	25.7	24.3	26.6	0.7	0.6	2024 · WDI
Homicides per 100,000	19.3	16.1	25.3	1.9*	n.a.	16.6	12.9	2024 · InSight Crime
PISA maths score	395	344	n.t.	343	n.t.	385	357	2022 · OECD PISA

WDI series NY.GDP.PCAP.PP.CD, IP.PAT.RESD, GB.XPD.RSDV.GD.ZS, SP.POP.SCIE.RD.P6, BX.TRF.PWKR.DT.GD.ZS, retrieved 31 Oct 2026 · *official, contested · n.a. no reliable series · n.t. not tested

The IDB projects we sized against

Project	Country, year	Amount	Benchmark for
CR-L1043 · Innovation and Human Capital for Competitiveness	Costa Rica, 2012	**US$35m**	Programme 1, capability agency
Border Integration Programme	Costa Rica	**US$100m**	Programme 2, corridor works
Border Integration Programme	Nicaragua	**US$55m**	Programme 2, corridor works
A recent programmatic policy-based loan	Mexico or Guatemala	From the IDB project database	Programme 3, budget support

IDB project database, project CR-L1043 · IDB Border Integration Programme documents

What the evidence says about formalisation

0.9% to 1.44%

OF GDP EVERY YEAR

Levy's estimate of the static efficiency cost of Mexico's formal-informal wedge (Levy, 2008).

0.34%

OF GDP, NET FISCAL COST

Antón, Hernández and Levy (2012, IDB): universal social insurance funded from general revenue, a uniform 16% VAT, and compensation for poor households.

Brazil's SIMPLES: three studies, three answers

Fajnzylber, Maloney and Montes-Rojas (2011)	**Large**
Monteiro and Assunção (2012)	**Limited**
Piza (2018)	**None**

Simplifying paperwork alone is not enough, so programme three removes the wage wedge itself.

Levy (2008), Good Intentions, Bad Outcomes, Brookings · Antón, Hernández and Levy (2012), IDB · Fajnzylber et al. (2011) · Monteiro and Assunção (2012) · Piza (2018)

Counterarguments we considered

“It is geography.”	Costa Rica shares the geography, and it is the outlier on almost every indicator on slide 3.
“It is corruption.”	It is a component of wall three, weak state capacity, rather than a separate cause.
“Nearshoring will fix it.”	Mexico's record 2025 FDI was 67.7% reinvested earnings and only 18% new investment, and assembly without linkages repeats the maquila pattern.

Secretaría de Economía, RNIE, 2025 FDI composition · slide 3 and slide 8 sources